

Social Media Management: (LinkedIn Automation System)

For the Official LinkedIn Page of Chhatrapati Shahu Ji Maharaj
University, Kanpur (CSJMU)

Project Documentation Report

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1. Brief Summary of the Project

The project titled Social Media Management: LinkedIn Automation System was developed to address a real, recurring communication problem faced by Chhatrapati Shahu Ji Maharaj University (CSJMU): maintaining a consistent, professional, and regularly updated presence on the university's official LinkedIn Page. The university regularly has achievements, events, projects, and academic milestones worth sharing, but writing a wellstructured, platform-appropriate post every time, and then publishing it with the correct images or video through LinkedIn's technical posting process, takes considerable time and effort.

To solve this problem, an AI-assisted LinkedIn content and publishing system was built specifically to manage CSJMU's official LinkedIn Page. The system takes a short description of a university event, achievement, or update from an authorized administrator or communications staff member, uses a Large Language Model (LLM) to draft a professional LinkedIn post in an appropriate tone and structure, allows the administrator to review and refine the draft, and then either publishes it immediately or schedules it for a future date and time. The system also manages media attachments (images or a single video), keeps a history of all posts, and supports multiple authorized users under a role-based account system, with the official CSJMU LinkedIn Page as the single, consolidated publishing destination.

The project was developed in two stages. The first stage was a working prototype built with Python and Streamlit, using an agentic workflow (built with LangGraph) that generates a post, evaluates it against a strict quality checklist, and automatically improves it if needed, before publishing it directly to the official LinkedIn Page. The second stage evolved this prototype into a complete full-stack web application (React frontend, Node.js/Express backend, and MongoDB database) with secure login, scheduling, drafts, history, and an administrative panel, so that the tool could be used on an ongoing basis to manage the university's LinkedIn presence rather than as a one-time script.

2. Background and Need of the Project

The official LinkedIn Page of a university is an important channel for institutional visibility, academic branding, and outreach. Regular, well-written posts about events, achievements, research initiatives, and student activities help build the university's reputation and keep its professional network informed and engaged.

In practice, however, maintaining this institutional presence is difficult for several reasons:

- Writing a post that is concise, professional, and free of factual errors takes time, especially when it has to be done repeatedly for different events and achievements.
- LinkedIn posts have their own conventions of tone, structure, and length that are different from a general description of an event. Converting raw event details into a ready-to-publish post requires a certain amount of content-writing skill.
- Publishing anything beyond plain text, such as an image or a video, through LinkedIn's official API involves a multi-step technical process (registering an upload, uploading the binary file, polling for processing status, and then referencing the resulting media asset in the post). This is not something a non-technical staff member can do directly.
- Posting consistently, on a regular schedule, requires manual reminders and discipline.

Without a mechanism to schedule posts in advance, gaps in the posting calendar are common.

- Since more than one staff member may be involved in drafting content for the official page, some form of account and access management is required so that publishing rights to the institutional page remain controlled and centralised.

The need, therefore, was for a tool that could take simple, informal event or achievement details as input, generate a publish-ready LinkedIn post automatically, allow an authorized reviewer to approve or refine it, and then handle the technical publishing and scheduling process for the official CSJMU LinkedIn Page without manual intervention every time.

3. Target Users of the Project

This project was built exclusively to manage the official LinkedIn Page of Chhatrapati Shahu Ji Maharaj University (CSJMU). It is not intended as a personal branding, posting, or profile-management tool for individual faculty members, assistant professors, staff, or students, and it does not connect to or publish on any individual's personal LinkedIn profile. All content generated and published through the system is attributed to, and appears on, the single official CSJMU institutional LinkedIn Page.

The intended users of the system are:

- The Communications / Social Media Team or other designated staff members responsible for maintaining CSJMU's official LinkedIn Page, who draft, review, and publish updates about university events, achievements, programmes, and announcements.
- University departments or administrative units, who can supply event or achievement details to be converted into ready-to-publish posts for the official page, subject to review and approval by the page administrator before anything is published.
- An administrator (superuser), who creates and manages the accounts of authorized content contributors, since public self-registration is intentionally disabled and every account must be created by an administrator.

Because the system is scoped to a single institutional LinkedIn Page, individual staff members are not given personal publishing access, and no personal LinkedIn account is ever connected or managed by the tool. This design keeps all content published under the CSJMU name reviewed and controlled through one consistent, institutionally-owned channel, rather than being distributed across separate personal profiles.

4. Problem Statement

The core problem addressed by this project can be stated as follows: there was no simple, self-contained tool that let an authorized university staff member describe an event or achievement in plain language and reliably obtain a professional, ready-to-publish post for CSJMU's official LinkedIn Page, complete with correctly attached media, published either immediately or at a scheduled time, without needing content-writing expertise or technical knowledge of LinkedIn's API.

The detailed aspects of the problem are:

- Raw event or achievement details are usually written informally and are not directly suitable for a professional LinkedIn audience.
- Manually rewriting this content into a LinkedIn-appropriate post, every time, is repetitive

and timeconsuming for the staff responsible for the official page.

- There was no automatic check to ensure that a drafted post stays accurate to the original details, keeps to an appropriate length, and follows a consistent institutional tone.
- LinkedIn's official posting API requires a specific, multi-step process for attaching images or video, which is impractical to perform manually for every post.
- Without scheduling, posts could only be published at the moment someone remembered to do so, leading to inconsistent posting frequency on the official page.
- Without user accounts and role separation, there was no controlled way to allow multiple staff members to draft content for the university page while keeping final publishing authority restricted.

5. Objectives of the Project

- To automatically generate a professional, well-structured LinkedIn post for the official CSJMU Page from a short, informal description of a university event or achievement.
- To evaluate the AI-generated post against defined quality criteria (accuracy, completeness, conciseness, tone, and structure) and automatically revise it if the criteria are not met.
- To allow an authorized human user to review, approve, or request changes to the generated post before it is published to the institutional page.
- To support multiple categories of official university posts (achievements, learning and outreach updates, project showcases, and more), each with its own tone and structure.
- To handle image and video uploads and publish them to the CSJMU LinkedIn Page using LinkedIn's official Assets and UGC Posts APIs.
- To allow posts to be saved as drafts, scheduled for a future date and time, or published immediately.
- To run scheduled publishing automatically in the background, without requiring a person to be present at the scheduled time.
- To provide secure, role-based access so that only an administrator can create new user accounts and manage other authorized contributors.
- To maintain a history of all drafted, scheduled, and published posts for the official page for future reference.

6. Proposed Solution

The proposed solution is an AI-assisted content generation and publishing system, delivered as a web application and dedicated entirely to managing CSJMU's official LinkedIn Page. It combines a Large Language Model (LLM) for drafting and refining posts with LinkedIn's official API for publishing, and adds scheduling, drafts, and user management on top so that the tool can be used continuously by authorized staff rather than as a single script.

6.1 Content Generation and Review Workflow

At the core of the system is a content-generation workflow, first built using LangGraph (an agentic workflow framework) and later re-implemented as a backend service. The workflow proceeds as follows:

- The authorized user selects a post category (for example, Institutional Achievement, Project Showcase, or Learning & Outreach) and enters a short description of the event or update, along with optional images or a video.
- If no title is supplied, a separate step automatically generates a suitable, professional event or post title from the description.
- The LLM drafts a first version of the LinkedIn post, guided by a prompt template specific to the chosen category, and by a fixed institutional context describing the university's tone, values, and communication style.
- An automatic evaluation step checks the draft against strict criteria — accuracy to the original details, completeness, length (a defined word-count range), professional institutional tone, and correct structure (headline, body, closing line, no hashtags or links in the body).
- If the draft fails any criterion, an optimisation step rewrites it based on the specific feedback, and the evaluation step runs again. This loop repeats for a limited number of iterations.
- Once the draft is approved (automatically or by the authorized reviewer), the reviewer previews it, and may accept it, ask for further improvement with their own feedback (which restarts the review loop), or discard it.
- On final approval, a closing step appends relevant hashtags (and any links found in the source details) without altering the approved text itself, producing the final publish-ready post.

6.2 Publishing and Scheduling Workflow

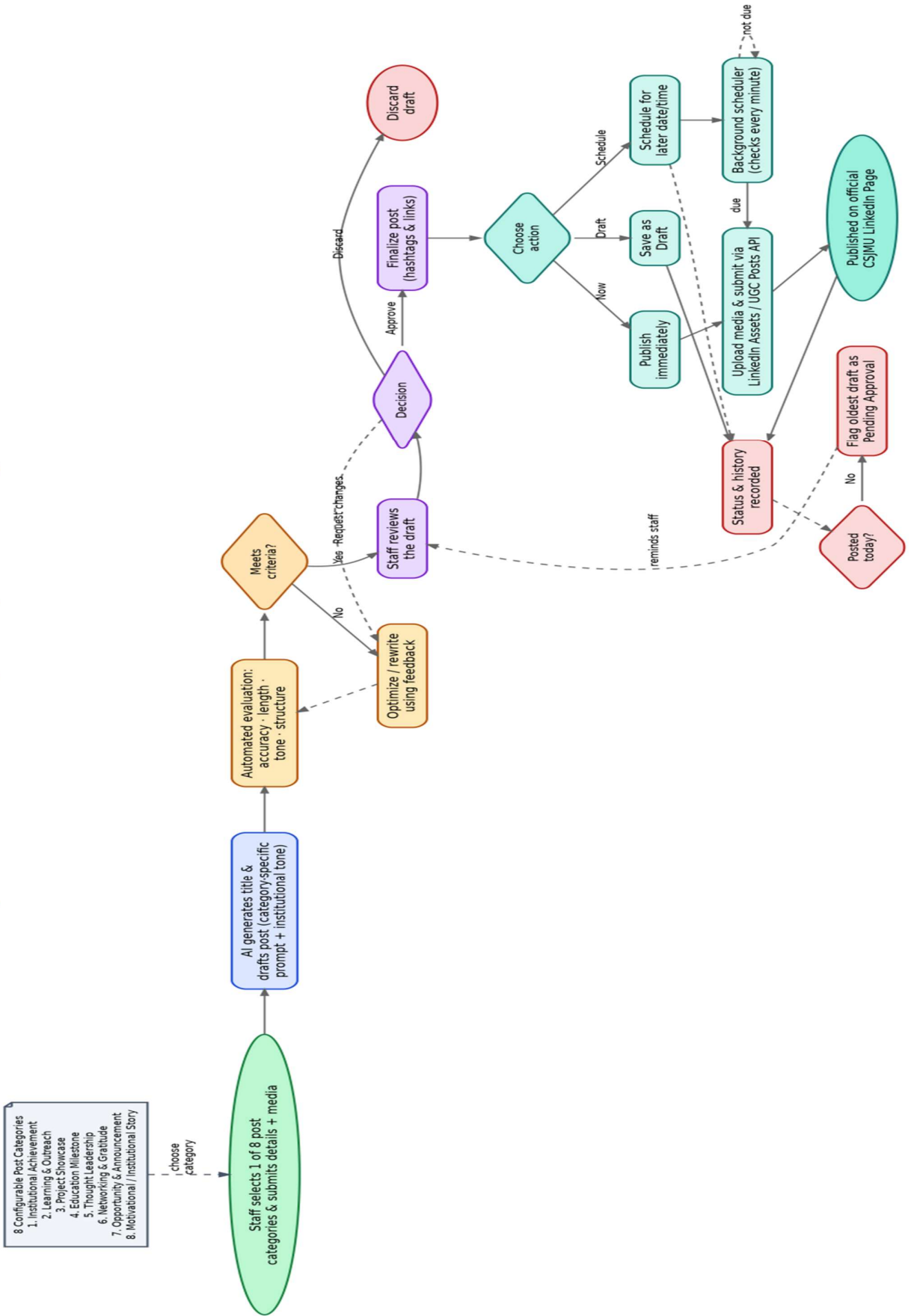
Once a post is finalised, the authorized user can choose to publish it immediately, save it as a draft, or schedule it for a later date and time. The publishing workflow is:

- The system identifies the CSJMU LinkedIn Page to publish under, using a securely stored access credential.
- Any attached images or the single attached video are uploaded through LinkedIn's asset-registration and upload process; for video, the system polls LinkedIn until the uploaded file is confirmed as processed and ready.
- The final post text, together with the resulting media references, is submitted to LinkedIn's UGC Posts API so it appears on the official CSJMU LinkedIn Page.
- For scheduled posts, a background job checks, once every minute, whether any post's scheduled time has arrived, and publishes it automatically to the official page when it is due. A daily backlog check looks at whether the official page has published or scheduled anything for the current day; if not, it automatically flags the oldest saved draft as pending approval so that the responsible staff member is reminded to keep posting regularly.

6.3 Complete Workflow — Flowchart

The diagram below summarises the complete, end-to-end workflow of the system on a single page — from the moment a staff member selects one of the eight configurable post categories and submits raw event or achievement details, through AI drafting and automated quality evaluation, human review, finalisation, and finally publishing or scheduling to the official CSJMU LinkedIn Page, including the background scheduler and the daily backlog reminder.

Complete Content Generation, Review, and Publishing Workflow



6.4 Overall Workflow Summary

In short, the complete workflow moves from raw event details, through AI drafting and automatic quality review, to authorized human approval, and finally to either immediate or scheduled publishing on the official CSJMU LinkedIn Page, with drafts, history, and reminders maintained throughout so that the university's LinkedIn presence stays active with minimal manual writing effort.

7. System Design and Components

The project consists of two implementations built one after the other: an initial Python-based prototype, and a complete web application that was developed from it. Each is described below.

7.1 Prototype (Python / Streamlit / LangGraph)

- `form.py` — The Streamlit-based user interface. It presents an event-submission form (event name, event details, and file upload for images or a single video), shows the AI-generated post on a preview page, and provides buttons to approve, request improvement, stop, or post the content.
- `linkedin.py` — Defines the core content-generation workflow as a state graph, with nodes for generating the post, evaluating it, optimising it based on feedback, capturing the reviewer's decision, and finalising it with hashtags. It uses few-shot example prompts and a structured (Pydantic-based) output format for evaluation results.
- `find_name.py` — A smaller, separate workflow that generates a professional event name/title automatically when the user does not provide one.
- `post_linkedin.py` — Handles direct integration with the LinkedIn API — retrieving the authenticated page's ID, uploading image or video assets, checking video-processing status, and finally submitting the post (text-only, image, or video) to the official LinkedIn Page.

7.2 Production Web Application (MERN Stack)

The prototype was rebuilt as a complete client-server web application with persistent storage, authentication, and scheduling, structured as follows:

Frontend (React, built with Vite):

- Login and Register pages for authenticated staff access.
- Dashboard for creating new posts by selecting a post category and entering details (see Figure 2).
- Drafts, Scheduled, and History pages for managing posts at each stage of their lifecycle.
- Settings page for managing the CSJMU LinkedIn Page access credentials and posting preferences.
- Superuser Panel for account administration.

Backend (Node.js with Express):

- `server.js` — Initialises the Express application, connects to MongoDB, serves uploaded media files, and automatically creates/synchronises a primary administrator (superuser) account from configuration values on every startup.
- `routes/auth.js` and `routes/posts.js` — Expose REST endpoints for login, listing available post categories, uploading media, and creating, updating, retrieving, and publishing posts.

- `models/User.js` — Stores each authorized contributor's username, hashed password (using `bcrypt`), role (user or superuser), account status, LinkedIn access credential used to publish to the official page, and backlog/auto-publish preferences.
- `models/Post.js` — Stores each post's category, event name and details, generated content, attached media files, status (draft, scheduled, published, or pending approval), scheduled time, resulting LinkedIn post ID, and a history of feedback used to refine it.
- `services/llm.js` — Wraps all calls to the LLM. It combines a fixed institutional tone context with one of the category-specific prompt templates to generate, evaluate, or refine post content, and includes a fallback mock mode for development or testing when no live model access is configured.
- `services/linkedin.js` — Provides the backend equivalent of the LinkedIn publishing logic — page lookup, media asset upload, and post submission.
- `services/scheduler.js` — Runs a recurring background job (every minute) that publishes any post whose scheduled time has arrived, and performs the daily backlog check described in the workflow section.
- `middleware/auth.js` — Verifies login tokens (JWT) on protected requests, confirms the account is active, and restricts certain actions to superuser accounts only.
- `config/postTypes.js` — Defines the supported official post categories (Institutional Achievement, Learning & Outreach, Project Showcase, Education Milestone, Thought Leadership, Networking & Gratitude, Opportunity & Announcement, and Motivational/Institutional Story), each with its own tone, target length, and prompt instructions.

Database:

- MongoDB is used to persist authorized user accounts and posts (including their status, schedule, and media references).

8. Technology Stack

Prototype:

- Python — core scripting language for the prototype.
- Streamlit — browser-based user interface for the prototype.
- LangGraph and LangChain — used to build the AI content-generation workflow as a graph of steps, with structured prompts and output parsing.
- Pydantic — used to define and validate the structured evaluation output produced by the LLM.

Production Web Application:

- React (with Vite) — frontend user interface.
- Node.js and Express.js — backend server and REST API.
- MongoDB with Mongoose — database and data modelling.
- JSON Web Tokens (JWT) and `bcrypt.js` — authentication and password security.
- Multer — handling of image and video file uploads.
- `node-cron` — background scheduling of due posts and daily backlog checks.
- Axios — HTTP communication with the LLM API and LinkedIn API.

AI / Language Model:

- A large language model (gpt-oss:120b) accessed through an Ollama-compatible cloud API, used for drafting, evaluating, and refining LinkedIn post content.

External Platform Integration:

- LinkedIn Assets API and UGC Posts API (v2) — used for uploading media and publishing text, image, and video posts to the official CSJMU LinkedIn Page.

9. Application Interface

The following screens illustrate the production web application as used by authorized CSJMU staff to generate, manage, and track official LinkedIn Page content.

9.1 Generate Posts Dashboard

From this screen, an authorized staff member selects one of the eight post categories (for example, Personal Achievement — Awards, internships, certifications), reviews the recommended tone and target length for that category, describes the achievement or event in the provided field, and triggers AI generation of a ready-to-publish LinkedIn post for the official CSJMU Page.

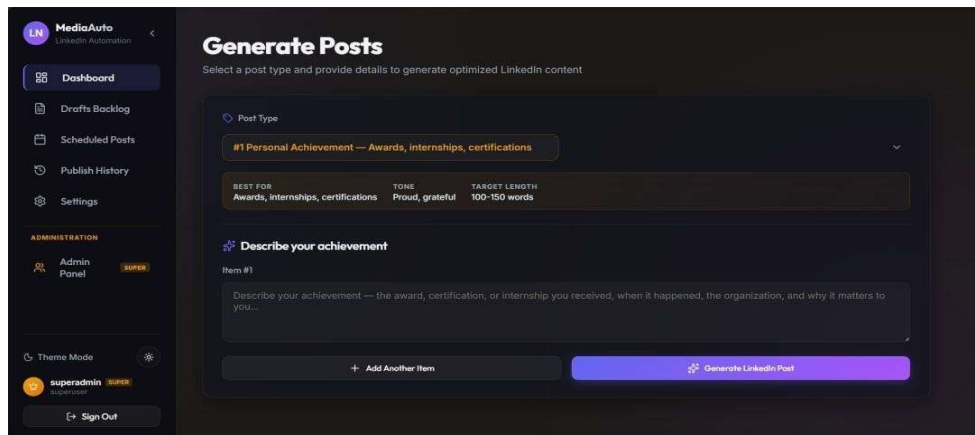


Figure 2: "Generate Posts" dashboard screen of the LinkedIn Automation System

9.2 Admin Panel

The Admin Panel, accessible only to superuser accounts, allows the administrator to create new contributor accounts, promote or demote roles, activate or deactivate users, reset passwords, and view at a glance how many accounts exist and are currently active — reinforcing the controlled, single-institutional-page access model described in Section 3.

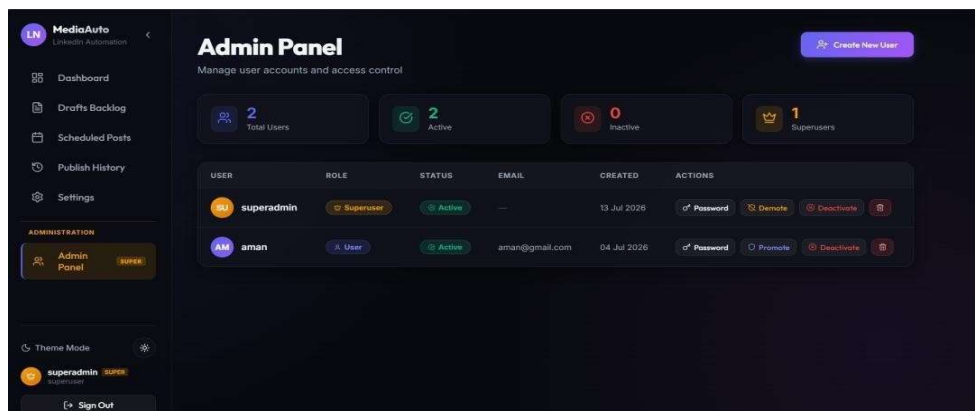


Figure 3: Admin Panel — user account and access management

9.3 Publish History

The Publish History screen provides a consolidated, filterable log of every post published to the official CSJMU LinkedIn Page across all authorized contributors, showing the category, author, publish timestamp, and a direct link to view the live post on LinkedIn.

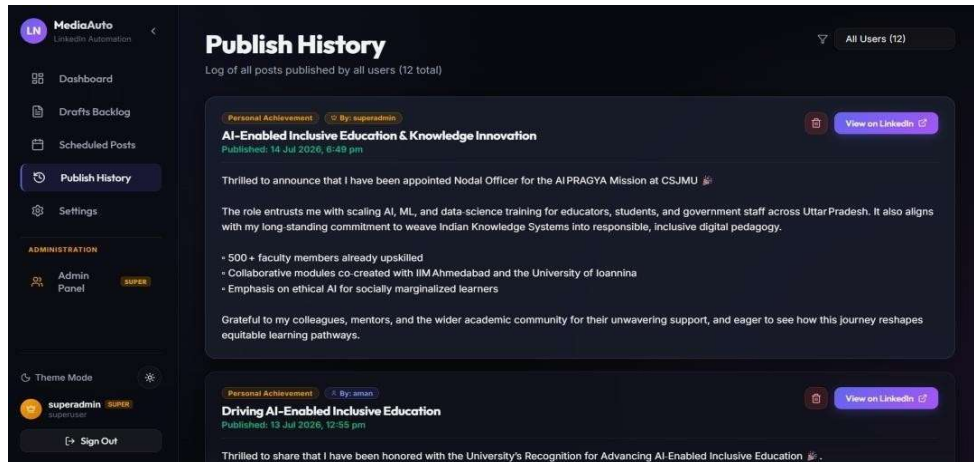


Figure 4: Publish History — consolidated log of posts published by all authorized users

10. Benefits of the Project

- Time saving: Drafting a professional LinkedIn post from raw event details, a task that could otherwise take significant time, is reduced to entering a short description and reviewing an AI-generated draft.
- Consistent quality: The built-in evaluation and refinement loop checks every draft for accuracy, completeness, length, tone, and structure before it reaches the reviewer, reducing the chance of publishing a poorly written or inaccurate post on the official page.
- Category-specific writing: Multiple distinct post categories, each with its own tone and structure guidelines, allow the system to produce content appropriate to very different institutional situations, from achievements to project showcases.
- Simplified media publishing: LinkedIn's technical, multi-step media-upload process is handled automatically, so the user only needs to attach a file rather than understand the underlying API.
- Reliable scheduling: Posts can be scheduled in advance and are published automatically by the background job, and the daily backlog check helps maintain a steady posting frequency on the official page.
- Controlled, multi-user access: Role-based accounts (user and superuser) and administrator-only account creation allow the tool to be used safely by more than one authorized staff member, while keeping all publishing consolidated under a single institutional page.
- Traceability: Every post's status, feedback history, and outcome are stored, giving a clear record of what was drafted, scheduled, and published on behalf of the university.

11. Additional System Capabilities

Beyond the core drafting, review, and publishing workflow, the system includes a set of supporting capabilities that give the authorized staff member full control over the daily backlog reminder, draft creation, and account access, described below.

11.1 Non-Intrusive Daily Reminder

If no post has been published or scheduled by 10:00 AM on a given day, the system surfaces a daily reminder listing the available drafts so that the official page does not go a day without an update. The reminder is entirely optional: the staff member can select any listed draft to Post Now, or dismiss the reminder with Skip / Decline Today and continue working immediately, without anything being published or saved in the background. The reminder never blocks or requires a posting action to proceed.

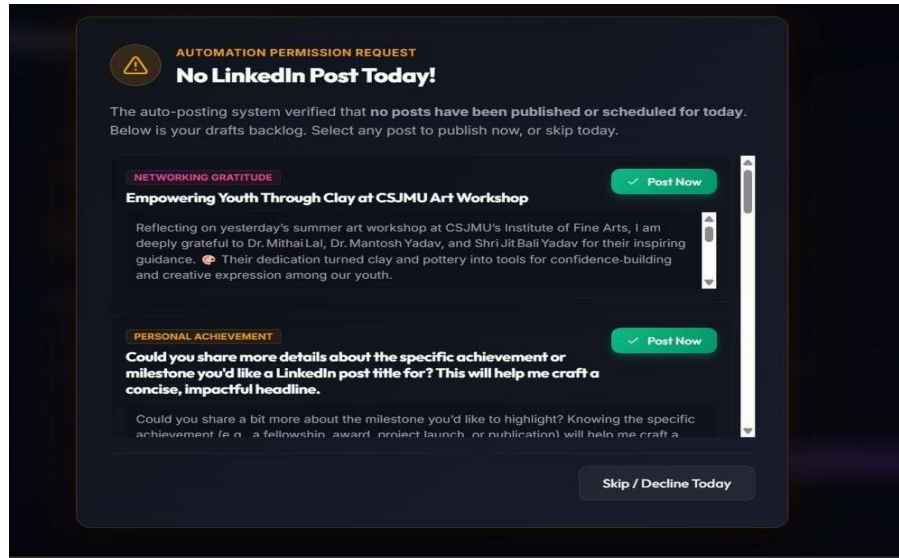


Figure 5: Daily reminder listing available drafts — Post Now on any item, or Skip / Decline Today

11.2 Multi-Post Backlog Review

The daily reminder (Figure 5) fetches every post in the drafts backlog that is eligible for scheduling and lists them together in a single scrollable panel, each with its category tag, title, and a short content preview. The staff member can review the full set at a glance and choose Post Now on any individual item, or close the panel without publishing anything. When the backlog is empty, the reminder displays a clear "nothing to publish" message instead of an empty panel.

11.3 Explicit Draft Saving

A newly generated post remains in the current working session and is added to the Drafts Backlog only when the staff member explicitly clicks Save Draft. This keeps the backlog limited to content the user has deliberately chosen to keep, rather than filling it automatically with every post generated during review and experimentation; refreshing the page or generating another draft without saving simply discards the unsaved version.

12. Limitations and Future Scope

12.1 Limitations

- The system depends on an external LLM API (accessed over the internet through an Ollama-compatible service); if this service or the configured API key is unavailable, content generation falls back to a limited mock mode rather than producing real AI-written content.
- The LinkedIn access credential for the official page is stored centrally and must be obtained and renewed manually, since an automatic OAuth login and token-refresh flow has not yet been implemented.
- The scheduler runs as part of the backend server process; scheduled posts are published only while the server is running, so an extended server outage could delay a scheduled post.

- The AI writing tone and context are currently configured for the single official CSJMU institutional voice; adapting the system for a different tone or department requires updating this configuration in the code.
- The system does not currently pull engagement data (likes, comments, views) back from LinkedIn, so post performance cannot be reviewed from within the application.

12.2 Future Scope

- Adding LinkedIn OAuth login with automatic token refresh, removing the need to manually configure access credentials for the official page.
- Supporting multiple configurable tone/content templates for different university departments, while keeping all publishing consolidated under the single official CSJMU LinkedIn Page.
- Bringing engagement analytics (likes, comments, impressions) back into the application for performance review.
- Extending the same content-generation and scheduling pipeline to other official university platforms, such as X (Twitter), Instagram, or Facebook.
- Hosting the scheduler and backend on a more resilient, always-on cloud environment to remove dependency on a single running process.
- Adding an institutional review/approval step before publishing, for use cases where posts must be checked by a communications team before going live.

13. Conclusion

This project set out to solve a practical problem for CSJMU: keeping a professional, well-written, and regularly updated official LinkedIn presence without spending large amounts of time writing and manually publishing every post. Starting from a Python-based prototype that combined an AI drafting-and-review workflow with direct LinkedIn API integration, the project was developed into a complete full-stack web application with secure accounts, drafts, scheduling, and an administrative panel — built exclusively to manage CSJMU's official LinkedIn Page.

The resulting system allows an authorized staff member to describe a university event or achievement in a few sentences and receive a publish-ready, professionally structured LinkedIn post, reviewed automatically for accuracy and tone, which can then be published immediately or scheduled for later, with images or video attached, and with LinkedIn's technical publishing steps handled entirely in the background. Overall, the project reduces the manual effort involved in social media management, helps maintain a consistent institutional presence on LinkedIn, and provides a foundation that can be extended with stronger authentication, analytics, and multi-platform support in the future.